



Designing modular capacitated emergency medical service using information on ambulance trip

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Abstract

In this paper, we investigate the design of a two-tiered emergency medical service (EMS) system. The objective remains in determining the location and the capacity of modular ambulance stations that minimize the EMS system's cost while respecting a pre-specified response time. A novel approach considering advanced information on ambulance trip and accounting for ambulance busy fractions is proposed. This approach is compared to its counterpart traditional approach that does not consider ambulance trip. Two mixed integer linear programs are developed. Experimentation of the two models was conducted on a real life case study. The obtained results pointed out the usefulness and superiority of the proposed approach. A cost saving of 3% is achieved in addition to the reduction in ambulance round-trip time.

Keywords Two-tiered EMS · Location allocation problem · Ambulance trip information · Modular capacity · Busy fraction · Mixed integer linear programming

Mathematics Subject Classification 90C11 · Mixed integer programming

1 Introduction

In this paper, we address the design of a two-tiered modular capacitated EMS system while taking into account advanced information on ambulance trip and ambulance busy fraction.

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Improving EMS system's performance means increasing patient survival probability while reducing pre-hospital time (Sudtachat 2014). The pre-hospital time is the ambulance travel time: from EMS station to incident scene in order to stabilize the patient then from the incident scene to the hospital where the patient is transferred. Evidently, the return time of ambulance to its base station or another station is not considered in pre-hospital time but should be accounted for as one of the determinants of ambulance availability particularly when ambulance needs to be disinfected in base station.

The pre-hospital time depends on several factors: ambulance base location, ambulance relocation and ambulance dispatching decisions. The main focus of this paper will be placed on ambulance location-allocation decisions that arise at the strategic-tactical level. Our goal is to determine the location of ambulance base stations, the number of ambulances to be assigned to each of them, and demand allocation to these stations. We present a solution approach for solving this location-allocation problem. Ambulance relocation and dispatching are operational decisions and are beyond the scope of this paper. At this level, it is worth noting that ambulance base stations and ambulance stations are used interchangeably.

While designing the EMS system, most researches consider explicitly only the response time (time between base location and demand zone). Obviously, any EMS system is aimed at providing rapid first aid to patients. In addition, given that the hospital where the patient would be transported is not part of the decision, the ambulance travel time and cost between the demand zone and the hospital are not affected by the retained configuration of the EMS system and hence the hospital (or patient's final destination) is not explicitly considered. Nevertheless, ambulance response time and service level depend on ambulance availability which conspicuously depends on ambulance trip time. Indeed ambulances become available when they go back to their base station after covering emergency (Van Essen et al. 2014; López et al. 2008).

The capacity of an ambulance station depends on both pre-hospital time and return time including the travel time from the demand zone to the hospital, in addition to the number of ambulances assigned to it. Given that the number of stationed ambulances limits the capacity of EMS facility, we consider different capacity levels. Capacity of each potential location must be chosen from a finite and discrete set of possible capacities (Correia and Captivo 2003). This problem is called modular capacitated location problem. Yin and Mu (2012) study the modular capacitated location problem for EMS considering the response time. They supposed that the maximal demand that could be served by one ambulance per year is known. However, this ambulance capacity is estimated while taking into account only the response time. The main goal of this study is to extend the existing works by explicitly considering the hospitals where patients have been admitted after being transported by the ambulance when designing a modular capacitated EMS system.

Traditionally, EMS systems use only one type of ambulances for all types of demand. However, recent researches point out that to improve the survival rate of life-threatening cases, it is important to cover this type of demand by first responder while waiting for the arrival of an ambulance (McLay 2009). Therefore, two types of vehicles should be used: a quick response vehicle and an ambulance. Nevertheless, it

is worth noting that when more than one type of vehicle is deployed by the EMS system, the problem is further complicated by whatever hierarchical relationship exists between the various types of vehicles (Tien et al. 1983). The relationship between different types of vehicles can be successively inclusive or successively exclusive. In a successively inclusive EMS hierarchy, one type of responder is capable to perform all procedures that other responder performs and additional life-saving procedures such as In Brazilian EMS (Silva and Pinto 2010) and Taipei EMS systems (Su and Shih 2003). In the successively exclusive EMS hierarchy, each type of responder provides different types of services such as in the Tunisian EMS system (Jebali et al. 2012). Henceforth, in this paper, we propose a location-allocation model for the design of a two-tiered “successively exclusive” EMS system. The objective is to find the configuration that minimizes the EMS system’s cost. The design of two-tiered EMS system has been considered by some studies (Mandell 1998; McLay 2009; Boujemaa et al. 2013, 2018). This research differs from earlier studies in respect not only of the approach but also of the objective. In this paper, we address the design of the EMS system, considering the patient’s final destination (the hospital) to express explicitly the modular capacity constraint while taking into account ambulance busy fraction. The objective is to minimize the total cost associated with the EMS system while respecting a pre-specified response time.

The remainder of this paper is organized as follows. Section 2 presents a review of previous models. Section 3 outlines ambulance location-allocation problem. A mixed integer linear programming (MILP) model is then presented. The proposed model is solved using branch and bound algorithm of CPLEX Optimization Studio 12.3 while considering real-world data of the northern region of Tunisia. In Sect. 3, computational results are presented and analyzed. Finally, in the conclusion, some perspectives of the current work are discussed.

2 Literature review

The ambulance planning problems have been widely investigated over the past 50 years. They treat operational, tactical and strategic decisions. In our review, the focus will be put on papers that either tackle ambulance location-allocation problems or deal with two-tiered EMS system or take into account the hospital (final destination of the patient) in ambulance planning.

The early ambulance location models are deterministic and most of them are extensions of the Location Set Covering Model (LSCM) introduced by Toregas et al. (1971) and the Maximum Covering Location Problem (MCLP) originated by Church and Reville (1974). The objective of LSCM is to minimize the number of ambulances needed to cover all the demands. MCLP aims at maximizing demand coverage through the best deployment of available ambulances. The first shortcoming to LSCM and MCLP models remains in the fact that they assume the EMS stations are uncapacitated. In other words, they assume that the ambulance is available all the day for any request. To overcome this shortcoming, (Yin and Mu 2012) addressed the MCLP while considering that an emergency facility could have different possible capacity levels (based on the number of stationed ambulances) called

the Modular Capacitated Maximal Covering Location Problem (MCMCLP). The authors propose two mathematical models for this problem: the MCMCLP-Facility-Constraint (MCMCLP-FC), in which the total number of stations to be located is fixed, and the MCMCLP-Non-Facility-Constraint (MCMCLP-NFC), in which, oppositely to the first model, the total number of stations is not fixed.

To increase the chance to have an ambulance available for an emergency, some researches suppose that a demand zone (also referred to as demand point) may be covered by more than one ambulance (Daskin and Stern 1981; Hogan and Revelle 1986). In Daskin and Stern (1981) and Hogan and Revelle (1986), the number of demand zones covered more than once is maximized. Another model called Double Standard Model (DSM) was proposed by Gendreau et al. (1997); it maximizes the demand covered twice within a given time standard. The authors developed a Tabu Search algorithm to solve this problem. In Doerner et al. (2005), the authors presented an extension of DSM where ambulance capacity considerations have been taken into account. They add a penalty term to the objective function to avoid the non satisfaction of capacity constraint. Similar to Yin and Mu (2012), the authors estimate the number of calls that can be covered by one ambulance. They used Ant Colony Optimization to plan the EMS stations in Austria and compared its performance to that of Tabu Search solution proposed in Gendreau et al. (1997). Their experiments revealed that Tabu Search could find better results with less computational effort, especially in large-sized problem instances.

Given that the capacity constraints in ambulance location-allocation models should prevent the assignment of an incoming call to a busy ambulance, the ambulance probability of being unavailable to cover a request, called busy fraction, has been introduced by Daskin (1983), Revelle and Hogan (1988), Beraldi et al. (2004) and Ingolfsson et al. (2008). Daskin (1983) proposed a model for the Maximum EXpected Covering Location Problems (MEXCLP). Later, several extensions of MEXCLP have been proposed. For instance, Revelle and Hogan (1988) developed the Maximal Availability Location Problem (MALP1 and MALP2) where the objective was to cover each demand with a given probability. Beraldi et al. (2004) proposed the use of joint probabilistic constraints to design a robust EMS system. The objective is to guarantee a high level of service by covering the random requests with a given probability. The proposed stochastic integer problem under probabilistic constraints was then reformulated and solved as a deterministic integer problem. In Ingolfsson et al. (2008), the authors tackled the problem of minimizing the number of ambulances required to ensure a service level while taking into account the uncertainty in delay, travel time and ambulance availability. In order to address stochastic aspects related to locating emergency services, some papers proposed the use of queuing models such as the hypercube queuing model (Larson 1974; Takeda et al. 2007) and priority queuing (Silva and Serra 2008).

To take into account the ambulance capacity, (Beraldi et al. 2004; Beraldi and Bruni 2009; Noyan 2010; Van Essen et al. 2014; Zhang and Jiang 2014) consider in their studies a planning horizon of 1 h. According to authors, an hour is a reasonable time for an ambulance to cover an emergency request and return to its base. Zuidhof (2010) investigated the variation of ambulance trip time in Amsterdam city; in other words, how long an ambulance will be occupied after its departure from the

ambulance station. Even though the travel time is generally less than 60 min, it can reach 100 min in some cases. He concluded that the ambulance trip time depends on urgency priorities, days of the week and different hours of the day.

Models presented above, consider only one type of ambulances. However, in some situations it is needed to dispatch more than one type of vehicles to the incident scene. It is the case of EMS systems that provide different types of services via different types of vehicles. Schilling et al. (1979) proposed the first models Tandem Equipment Allocation Model (TEAM) and Facility-Location, Equipment-Emplacement Technique (FLEET) considering two types of vehicles for fire companies: basic vehicles and specific vehicles while maximizing the coverage. The (TEAM) supposed that specific equipment could only be located in tandem with basic equipment. In the (FLEET) model, both equipment types can be either located in tandem or singly. These models were tested in a study of the Baltimore City Fire Protection System. Marianov and Revelle (1992) proposed a probabilistic model of emergency system with two types of ambulances in which a reliability requirement is imposed on the availability of servers. Mandell (1998) proposed a probabilistic covering-type model for two-tiered EMS systems not nested, i.e., it is not necessary to restrict the sites of second responder units to those sites where first responder unit is located. Su and Shih (2003) proposed a simulation model to evaluate and perform two-tier rescues system in Taipei, Taiwan. McLay (2009) develops an integer programming model MEXCLP2 (extension of MEXCLP) while considering two types of medical units. Results were illustrated using real-world data collected from Hanover County, Virginia. (Boujemaa et al. 2018) study a two-tiered EMS system under demand uncertainty. They developed a two-stage stochastic program with recourse to deal with demand uncertainty. The proposed model is then solved using the sample average approximation algorithm. Similar to (Beraldi et al. 2004), the capacity constraint is implicitly taken into account while considering a horizon of 1 h. However, 1 h is an estimate of ambulance service time. That is why, authors conduct a sensitivity analysis in order to investigate the behavior of the system considering a service time varying between 45 and 75 min.

As it can be noticed, all the aforementioned papers considered only the response time or supposed that the total ambulance travel time cannot exceed 1 h (Beraldi et al. 2004; Noyan 2010; Zhang and Jiang 2014; Van Essen et al. 2014; Boujemaa et al. 2018). Moreover, in all these works, the ambulances are considered available when they return to their base station, possibly after a cleaning process. Van Essen et al. (2014) develop a strategic-tactical ambulance planning model while taking into account the hospital. The availability of ambulance is accounted for through its total travel time (travel time from base station to patient's location + treatment time + travel time from patient's location to hospital + time spent in the hospital + travel time from hospital to base station + cleaning ambulance's time). Thereafter, for the sake of simplification, a horizon of 1 h is adopted. The authors considered that 1 h is a reasonable estimate of ambulance total travel time.

Some recent papers tackle the ambulance relocation/redeployment problem while considering the final destination of the patient (López et al. 2008; Sudtachat 2014; Sonoda and Ishibai 2015; Maleki et al. 2014). According to (López et al. 2008), response time has a significant effect on ambulance dispatching problem. They

proposed two algorithms for dispatching ambulances: (1) considering only response time and (2) considering response time and return time. The authors showed that ambulance availability is increased when dispatching considers the total ambulance travel time. Sudtachat (2014) developed a discrete event simulation model of an EMS system to study the performance of different dispatching policies. The state of ambulance is “busy” when it is in service and providing patient transport to a hospital. Its state becomes again “idle” as soon as it returns to its original station.

The literature discussed above shows that only some papers study the design of two-tiered EMS system. However, as far as the capacity constraints are concerned, the existing works do not consider explicitly the total ambulance travel time. In this paper, we propose a novel location-allocation model to design two-tiered EMS system, considering at the same time: total service time, busy fraction and modular capacity of base stations. We focus on strategic-tactical planning of EMS system. The following questions are tackled: what are the EMS stations to open? How many ambulances of each type to assign to each station? What are the demand zones to be covered by each station?

3 Model formulation

3.1 Problem description

The proposed model is based on the Tunisian EMS system, SAMU (*Service d'Aide Médicale Urgente*). The SAMU is the coordinator and the regulator of pre-hospital healthcare services provided by medical teams responsible for providing first aid and patient transport to the hospital, SMUR (*Service Mobile d'Urgence et de Réanimation*). Each SAMU covers the demand of a Region containing several cities. When an emergency call is received by SAMU's phone center, a regulating doctor states the priority of the emergency call. Four degrees of priority can be assigned to a call: (1) no emergency case (code 1); (2) only need for transportation to the hospital (code 2); (3) routine incident (code 3) and (4) emergency case involving danger to human life (code 4). Obviously, calls of code 4 deserve higher priority over all other calls. Calls of code 1 and 2 do not need however any intervention of the SMUR, these calls are oriented to the appropriate services. For calls of code 3 and 4, at least an ambulance crew composed of a doctor, a nurse and a driver and an ambulance adequately equipped are dispatched from one EMS station to the incident scene. Two kinds of vehicles could be used: (1) ambulance equipped with Basic Life Support (in the paper they are referred to as BLS) and (2) ALS non-transport Quick Response Vehicles (QRVs). An emergency case of code 3 needs a BLS ambulance; however, an emergency case of code 4 requires the intervention of both a QRV and a BLS. The QRV transports rapidly the medical team to the incident scene to start providing first aid and then a BLS arrives and transports the patient to the hospital. After serving a request, the ambulance is supposed to return to its home base (the original EMS station).

In some countries such as United States, Canada, United Kingdom and Italy there are regulations specifying standard response times (Takeda et al. 2007). In Tunisia,

no specific regulations regarding response time are explicitly stated. Nevertheless, in the SAMU, it is commonly agreed that the response time must not exceed 20 min for a request of code 3 and 15 min for a request of code 4. SAMU-SMUR personnel strive to respect this rule.

The emphasis of this paper is placed on investigating EMS system configuration at the strategic-tactical level. The objective is to minimize the overall costs while respecting the pre-specified standard response time. We propose an approach that takes into account ambulance trip, busy fraction and modular capacity constraints.

To estimate the common busy fraction (σ_k) for vehicles of type k , we use the formula developed by (Daskin 1982): the mean number of daily hours of service of vehicle type k needed by the system divided by the number of daily hours available, assuming that p_k servers of type k will be used. In this paper, p_k presents the maximum number of vehicles of type k that can be offered by the Ministry of Health.

Concerning the ambulance trip, five interval times are considered: (1) the response time (time between the call and the ambulance arrival on the scene); (2) the on-scene time (the time from ambulance arrival to the scene to the time when the ambulance departs to the hospital); (3) the transport time (from when the ambulance departs from the scene to the time when the ambulance arrives to the hospital); (4) the drop-off time (the time needed to discharge the patient and process his admission to the hospital); (5) the return time (time from the hospital to ambulance home base). Using appropriate forecasting models, the yearly demand of each geographical zone that will be taken over by a given hospital of the SAMU Region is estimated. Then, we consider detailed data on demand including the final destination of the patient and time intervals, most of which are considered to be part of pre-hospital time.

EMS system performances are assessed and compared to those when the patient's final destination is not considered. The objective is to investigate the relevance of taking into account patient's final destination in EMS system configuration at the strategic-tactical level.

The model tries to minimize the EMS system's cost while considering modular capacity constraints. This cost includes the cost of opening an EMS station, the cost of allocating resources to the opened EMS stations and the transportation costs. Each EMS station has a capacity which depends on the number of the allocated ambulances and their busy fraction. There is a set of demand points, at which requests are generated, and a set of potential locations, where EMS stations might be opened. Ambulances and SMUR teams are assigned to EMS stations. A maximum number of vehicles of each type is considered, then ambulance allocation to EMS stations will specify the number of needed ambulances to cover demand points. In addition, the following assumptions are considered: (1) one facility could cover several demand points, (2) each demand point could be covered by more than one site.

3.2 Notation

Sets

M	set of demand points
N	set of potential EMS stations
H	set of hospitals
L	set of demand types
T	set of vehicle types

Indices

$i \in M$	index of demand points
$j \in N$	index of EMS stations
$h \in H$	index of hospitals
$l \in L$	index of demand types
$k \in K$	index of ambulance types

Parameters

R_k	maximum response time required to cover demand needing a vehicle of type k
δ_{jik}	travel time between potential EMS station j and demand point i when covered by ambulance of type k
β_{ihk}	travel time between demand point i and hospital h covered by ambulance of type k
γ_{hjk}	travel time between hospital h and potential EMS station j by an ambulance of type k
Ts_k	on-scene time of ambulance of type k
Ta_{hk}	drop-off time of ambulance of type k in hospital h
p_k	number of ambulances of type k
C_k	yearly capacity of vehicle of type k (in h)
D_{ilh}	number of requests of type l received from demand point i including first care medical assistance and transportation to hospital h
Ψ, Ω	two big numbers
v_k	cost of assigning one ambulance of type k to one ambulance station (<i>in TND</i>); it includes ambulance amortization cost and the compensation of ambulance crew
w_{ijk}	cost of using ambulance of type k from ambulance station j to demand point i (<i>in TND</i>)
s_{jkh}	cost of using ambulance of type k from hospital h to ambulance station j (<i>in TND</i>)
f_j	cost of opening ambulance station j (<i>in TND</i>); it includes equipment and furniture cost, and the compensation of ambulance crew
Δ_{kl}	$= 1$, if an ambulance of type k is required to cover demand of type l

- α_k = 1, if an ambulance of type k returns to the ambulance station (= 1 for QRV),
 = 0, if an ambulance of type k transports patient to the hospital (= 0 for BLS)
 σ_k busy fraction of vehicle of type k

Decision variables

- X_{ijk} = 1 if demand point i is covered by a vehicle of type k located in station j ; 0 otherwise
 Y_j = 1, if station j is opened; 0 otherwise
 Z_{jk} number of ambulances of type k assigned to station j
 N_{ijlh} number of requests of type l incoming from demand point i , covered by a vehicle from station j and including transportation to hospital h

3.3 Mathematical model (P1)

The proposed Mixed Integer Linear Program (P1) tries to find out the design of the EMS system that minimizes the system cost for a predefined response time. The total cost is composed of, the cost of opening vehicle stations, the cost incurred by vehicle assignment to opened stations and, the travelling cost. The MILP can be modeled as follows:

$$\begin{aligned} \text{Min} \left[\sum_j f_j Y_j + \sum_k \sum_j v_k Z_{jk} + \sum_i \sum_j \sum_k \sum_l \sum_h (\Delta_{kl}(1 + \alpha_k)) w_{ijk} N_{ijlh} \right. \\ \left. + \sum_i \sum_j \sum_k \sum_l \sum_h s_{hjk} (1 - \alpha_k) N_{ijlh} \right] \end{aligned} \quad (1)$$

Subject to

$$X_{ijk} \delta_{jik} \leq R_k \quad \forall i, j, k \quad (2)$$

$$\sum_j Z_{jk} \leq p_k \quad \forall k \quad (3)$$

$$\sum_k X_{ijk} \leq \sum_h \sum_l N_{ijlh} \quad \forall i, \forall j \quad (4)$$

$$\Delta_{kl} \sum_h N_{ijh} \leq \Psi X_{ijk} \quad \forall i, j, k, l \quad (5)$$

$$\sum_i \sum_k X_{ijk} \leq \Omega Y_j \quad \forall j \quad (6)$$

$$\sum_i \sum_k X_{ijk} \geq Y_j \quad \forall j \quad (7)$$

$$\begin{aligned} & \sum_i \sum_l \sum_h N_{ijlk} [\delta_{jik} \Delta_{kl} (1 + \alpha_k) + Ts_k + (1 - \alpha_k) (\beta_{ihk} + Ta_{hk} + \gamma_{hjk})] \\ & \leq C_k Z_{jk} (1 - \sigma_k) \quad \forall j, k \quad (8) \end{aligned}$$

$$\sum_j N_{ijlh} \geq D_{ilh} \quad \forall i, \forall l, \forall h \quad (9)$$

$$Y_j, X_{ijk} \in \{0, 1\} \quad \forall i, j, k \quad (10)$$

$$Z_{jk} \text{ integer} \quad \forall j, k \quad (11)$$

$$N_{ijlh} \geq 0 \quad \forall i, j, l, h \quad (12)$$

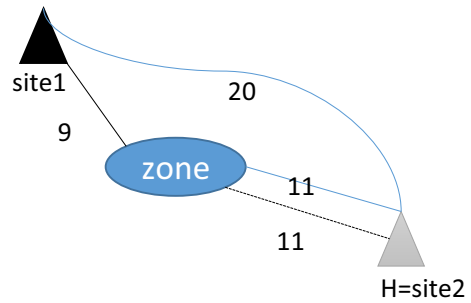
Constraints (2) guarantee the respect of the maximum response time allowed for the coverage of each demand point. Constraints (3) restrict the total number of ambulances. Constraints (4) and (5) state that a demand point i is covered by a vehicle of type k located at site j if and only if there are requests of type l (needing vehicle of type k), of demand point i and covered by site j . Constraints (6) and (7) indicate that a site is opened when it covers at least one demand point. The family of constraints (8) permits to respect site capacity. The site capacity is adjusted by ambulances busy fraction. By considering integer values of the demand, constraints (9) with the objective function ensure the coverage of requests of each demand point. Constraints (10), (11) and (12), respectively, specify binary, integer and positive decision variables.

4 Experimentation and results

In this section, first we present a small sized example that demonstrates the interest of considering total ambulance trip in EMS system design. Then, we provide meaningful results and insights derived from a real life case study.

4.1 Illustrative example

Let us consider the EMS system depicted in Fig. 1. It consists of one demand zone, two potential locations for ambulance stations (site 1 and site 2) and one hospital H (located at site 2). For the sake of simplicity, let us assume that all emergency calls coming from this demand zone are of code 3 and that all patients must be transferred to hospital H by a BLS ambulance. Table 1 reports ambulance travel time (in min)

Fig. 1 Illustrative example**Table 1** Ambulance travel time (in min)

	Demand zone	Hospital
Site 1	9	20
Site 2	11	0

between each ambulance station and the demand zone and between the demand zone and the hospital H.

Traditionally, we locate ambulance station at the nearest site to the demand zone, namely site 1. The time between the demand zone and the hospital is 11 min (min), the on-scene time is 10 min, the drop-off time is 10 min and the setup time (cleaning) is 10 min. Hence whenever an ambulance departs from site 1 to serve an emergency request it will become available again after $(9 + 10 + 11 + 10 + 20 + 10)$ 70 min. However, if ambulance station is located at site 2, the ambulance will become available again after $(11 + 10 + 11 + 10 + 0 + 10)$ 52 min.

Furthermore, while assuming that ambulance station is located at site 2, we investigate the difference between (1) our approach considering information on ambulance trip and (2) the traditional approach that does not explicitly consider ambulance trip but rather assumes that the vehicle is available after 1 h (Beraldi et al. 2004; Noyan 2010; Zhang and Jiang 2014; Boujemaa et al. 2018). As it can be noticed, according to our approach, the ambulance is available after 52 min if a cleaning process is needed and after 42 min if not. Hence in these works the number of required vehicles might not be accurately estimated.

4.2 Case study

Our experimentation is based on real-life data collected from the Tunisian Ministry of Public Health over the activity of EMS system of the Northeast Region of Tunisia (referred to as, SAMU01) during 2010. The considered Region includes seven governorates, has an area of 12,075 km² and its population amounts to a total of 3.2 million inhabitants, which corresponds to 37% of Tunisia's population (according to 2013 estimates of the Tunisian Statistical Institute). The SAMU01 is the coordinator and the urban regulator of the different EMS stations of the aforementioned

governorates. Only requests of code 3 and 4 are taken into account as they trigger the use of an ambulance. The demand points are represented by the centers of the seventy eight delegations of the considered governorates ($|M|=78$). The thirty one hospitals, health centers and existing vehicle stations located in the Region are deemed as potential sites for vehicle stations ($|N|=31$). Each demand point can be covered while respecting the predefined response time at least by one of the potential sites. In the Region, twenty three hospitals ($|H|=23$) could admit emergency patients. For each demand point, we know the set of hospitals to which the patient may be transported after receiving first aid care. In addition, historical data on the activity of EMS system could be used to forecast the number of requests of one demand point covered by each of these hospitals (Ingolfsson et al. 2003; Schmid 2012; Sudlow et al. 2011; Patel et al. 2012). In the dataset provided, 12,401 requests were answered by SAMU01; 91% of them were of code 4. However, this number of requests does not reflect the total number of emergencies of the Northeast Region of Tunisia. There are emergencies that are taken over by other pre-hospital service providers, such as the firefighters or private hospitals and clinics.

The time between different points is determined based on the distance and average vehicle speed between each two locations. The QRV vehicle is more speedy than BLS ambulance (the speed of BLS is assumed to be approximately equal to $0.85 \times$ the speed of QRV). The cost is supposed to depend on the distance and is equal to 1.2 TND/km for BLS and 0.8 TND/km for QRV (this cost is based on fuel consumption of each vehicle type). The average on-scene time is respectively, 15 and 20 min for the BLS and QRV ambulances. The average drop-off time in the hospital is 30 min. The capacity of one vehicle is expressed in minutes. Each vehicle could be used 24 h a day over 350 days of the year.

The common busy fractions of each type of vehicles are estimated to 30% considering a maximum number of 14 vehicles (for each type QRV and BLS).

To assess the performance and the interest of the proposed approach, we compare it to its counterpart approach that does not consider the entire ambulance trip in the design of the EMS system. The latter uses a MILP (P2) very similar to (P1). Indeed, only the objective function (1) and constraints (8) should be changed. They are replaced by (13) and (14), respectively.

$$\text{Min} \left[\sum_j f_j Y_j + \sum_k \sum_j v_k Z_{jk} + \sum_i \sum_j \sum_k \sum_l \Delta_{kl} w_{ijk} N_{ijlh} \right] \quad (13)$$

$$\sum_i \sum_l \sum_h N_{ijk} [\delta_{jik} \Delta_{kl} (1 + \alpha_k) + Ts_k + (1 - \alpha_k) (RT + Ta_{hk})] \leq C_k Z_{jk} (1 - \sigma_k) \quad \forall j, k \quad (14)$$

where RT is a constant used as an upper bound of, the sum of the travel time between the demand point and the hospital and, the travel time between the hospital and EMS station. Based on the provided dataset, RT was given the value of 70 min.

First, 30 problem instances have been tested. They encompass 10 scenarios based on the provided real-life data. Each scenario is obtained by multiplying the annual number of requests by a coefficient varying from 1 to 10. This could be justified by a

benchmark established while comparing the number of requests per 100 persons over 1 year of different regions of other countries of the world. We noticed that the registered number of requests per 100 persons in the northeast region of Tunisia is relatively low (0.39 requests per 100 persons in 2010) with comparison to other countries. For example, in the southern region of Ireland, the number of incidents by 100 persons reached 3.3 in 2006 and this number was expected to increase in the upcoming years.¹ Besides, the number of incidents in Tunisia has witnessed a noticeable increase over the last 3 years. It is clear that the the maximum budget allocated for the deployment of ambulances must be increased in order to meet the demand, when the requests increase. The busy fraction is assumed to 30% for each scenario.

The MILPs (P1) and (P2) generated by these scenarios are solved with IBM ILOG CPLEX 12.3 on a PC Pentium IV, 3.0 GHz. In order to gauge the trade-off between the EMS system cost and the service level, for each of these 10 basic scenarios, the following three response time standards have been, respectively, considered for BLS and QRV ambulances: 20–15 min; 25–20 min and 30–25 min. MILPs (P1) comprise 116,158 variables and 131,930 constraints. MILPs (P2) comprise 9766 variables and 22,106 constraints. All MILPs generated by these scenarios have been solved in a reasonable computational time [less than 40 min for (P1) and less than 10 min for (P2)].

For all scenarios we find out the EMS system configuration by, first, solving (P1), then by solving (P2). In order to compare the two configurations, we run (P1) while considering the EMS stations proposed to open according to (P2). Indeed, this decision could not be changed after some months because the costs of closing and reopening EMS stations are relatively high (Basar et al. 2011). Nevertheless, the number of ambulances to be located in each EMS station could be revised.

Table 2 presents some computational results. For each scenario, the following information is provided: the problem instance number (N°), the considered Demand Level or coefficient (DL), the considered Response Time limit (RK), the value of (P1) objective function (Z_AFD), the value of (P2) objective function (Z_WAFD), the EMS system cost if (P2) solution is implemented (Z_P1WAFD), the gap between the two configurations in terms of, total cost (GAP_Z), transportation cost (GAP_TC) and BLS vehicle time (GAP_ARAT). These gaps are expressed in percentage and are determined as follows:

$$GAP_Z = \frac{Z_P1WAFD - Z_AFD}{Z_AFD} * 100 \quad (15)$$

$$GAP_TC = \frac{TC_P1WAFD - TC_AFD}{TC_AFD} * 100 \quad (16)$$

$$GAP_ARAT = \frac{ARAT_P1WAFD - ARAT_AFD}{ARAT_AFD} * 100 \quad (17)$$

¹ Demand Analysis and Tactical Deployment of Ambulance Services in the National Ambulance Service—Southern Region, Report for the Pre-Hospital Emergency Care Council and the National Ambulance Service, *Feidhmeannacht na Seirbhíse Sláinte-Health service executive*, July 2010.

Table 2 Computational results

N°	DL	RK	Z_AFD	Z_WAFD	Z_PIWAFD	GAP_Z	GAP_TC	GAP_ARAT
1	1	20–15	3,609,538	3,258,834	3,610,244	0.02	0.97	0.24
2		25–20	2,782,296	2,459,844	2,785,523	0.12	0.83	0.79
3		30–25	2,647,138	2,265,518	2,647,138	0.00	0.00	0.00
4	2	20–15	4,186,188	3,614,201	4,186,188	0.00	0.00	0.00
5		25–20	3,653,196	2,953,262	3,671,056	0.49	26.13	18.09
6		30–25	3,570,085	2,827,356	3,649,549	2.23	27.93	20.28
7	3	20–15	4,905,012	4,104,926	4,967,722	1.28	15.35	11.11
8		25–20	4,373,987	3,561,350	4,473,482	2.27	16.77	13.06
9		30–25	4,276,439	3,372,706	4,402,043	2.94	23.79	16.28
10	4	20–15	5,543,201	4,661,190	5,682,408	2.51	15.31	11.12
11		25–20	5,104,018	4,094,484	5,371,746	5.25	27.11	18.59
12		30–25	5,016,625	4,080,548	5,355,482	6.75	34.41	23.05
13	5	20–15	6,295,155	5,140,965	6,510,858	3.43	15.53	11.17
14		25–20	5,776,867	4,693,531	6,212,778	7.55	21.90	16.45
15		30–25	5,742,518	4,633,379	6,084,641	5.96	25.03	17.80
16	6	20–15	7,088,524	5,720,440	7,481,405	5.54	15.79	11.87
17		25–20	6,611,233	5,342,827	7,033,962	6.39	17.59	13.62
18		30–25	6,441,300	5,232,273	6,968,887	8.19	21.40	15.55
19	7	20–15	7,912,847	6,278,883	8,362,188	5.68	15.28	10.81
20		25–20	7,334,847	5,999,435	8,102,230	10.46	20.72	16.48
21		30–25	7,246,924	5,933,690	7,367,659	1.67	5.93	3.93
22	8	20–15	8,590,364	6,972,376	9,189,157	6.97	15.31	11.12
23		25–20	8,187,621	6,591,333	8,218,254	0.37	1.07	0.96
24		30–25	7,988,680	6,521,267	8,011,219	0.28	0.63	0.61
25	9	20–15	9,246,478	7,562,753	9,262,158	0.17	−1.40	−0.86
26		25–20	8,853,250	7,219,401	8,897,590	0.50	3.19	2.46
27		30–25	8,727,617	7,098,094	8,971,119	2.79	5.62	4.83
28	10	20–15	9,969,517	8,161,119	9,969,517	0.00	0.00	0.00
29		25–20	9,683,951	7,813,642	9,726,588	0.44	1.08	1.02
30		30–25	9,580,391	7,705,958	9,595,888	0.16	0.51	0.59

where TC_PIWAFD and ARAT_PIWAFD, respectively, denote the incurred transportation cost and, the BLS vehicle utilization time, of the EMS system configuration obtained when patient's final destination is considered.

TC_AFD and ARAT_AFD, respectively, denote the incurred transportation cost and the BLS ambulance utilization time of the EMS system configuration obtained when patient's final destination is not considered.

In 90% of cases, a cost cutting is achieved whenever patient's final destination is considered in the design of the EMS system. On average, a cost cutting of 3% is achieved. This cost cutting is remarkably low (less than 2%), particularly for low (1 and 2) and high (9 and 10) demand coefficients. For a low demand, the

transportation cost is lower than the other terms of the objective function in both models. (P1) could propose to open more EMS stations than what proposes (P2). Indeed, transportation cost in (P1) has a higher weight than in (P2), and this could create a room to optimize the total cost by opening EMS stations to be closer to the demand zone and the hospital. Opening a larger number of EMS stations incurs costs that hide the gain in transportation cost. The gap in the total cost could be low but there is a substantial gain in term of transportation cost as we can notice, for example, in problem instance 5. When the demand is relatively high, transportation cost, in both models, are becoming much larger than the other cost terms. In order to optimize transportation cost, both (P1) and (P2) tend to propose to open larger number of EMS stations. This leads to reduce the gap between the two configurations because the larger number of EMS stations opened, the greater likelihood to be close to the demand zone and hospital. An average of 6% cost cutting has been noticed for scenarios with a demand coefficient of 5, 6 and 7. This cost cutting reaches 10.46% for problem instance 20.

For more in-depth analysis of the obtained results, in Table 3, we present additional details on EMS system configurations proposed by (P1) and (P2) for scenarios with a demand coefficient (DL) of 1, 5 and 10 [corresponding to problem instances (1, 2, 3), (13, 14, 15) and (28, 29, 30)]. For each problem instance, the following information is provided: number of opened stations as proposed, respectively, by P1 and P2 solutions (OS_P1, OS_P2), number of ambulances of type QRV needed by the system as proposed, respectively, by P1 and P2 solutions (QRV_P1, QRV_P2), number of ambulances of type BLS needed by the system as proposed, respectively, by P1 and P2 solutions (BLS_P1, BLS_P2).

For scenarios with a demand coefficient of 1 and 10, there is a slight difference between (P1) and (P2) solutions regarding the stations proposed to be opened. For problem instances 3 and 28, the same stations have been proposed by both models. However, the two models, except in problem instances 1 and 3, propose different number of ambulances for the EMS system noticeably when the demand level increases. In scenario 5, P1 proposes to open more vehicle stations than P2. Even if this incurs higher opening costs, an average total cost cutting of 5.6% is achieved, besides an average transportation cost reduction of 20.8%. Nevertheless, (P2) tend to overestimate the number of BLS ambulances needed by the system because of the capacity constraints (13). These constraints consider an upper bound of, the sum of the travel time between the demand point and the hospital and, the travel time between the hospital and EMS station, because the final destination of the patient transported by the BLS ambulance is not taken into account. Naturally, the number of BLS ambulances is overstated, and more significantly, when the demand increases. The number of QRV ambulances is the same or slightly higher in (P1) configuration even though capacity constraints (8) and (13) in (P1) and (P2), respectively, are similar regarding the dimensioning of this resource. Indeed, (P2) tries to locate stations in order to minimize the transportation cost without accounting for patient's final destination. This incurs the opening of vehicle stations closer to the demand zones with comparison to (P1).

As it can be observed from Table 3, according to the EMS system configuration proposed for scenario 1 with a demand coefficient 1 and response time standards

Table 3 Comparison of P1 and P2 configurations

DL	OS_P1	OS_P2	QRV_P1	QRV_P2	BLS_P1	BLS_P2
1	13 (2, 3, 16, 22–31)	13 (2, 3, 13, 22–31)	13	13	13	13
	9 (3, 21, 22, 23, 25, 28–31)	9 (3, 22, 23, 25, 27–31)	9	9	9	10
	8 (16, 22, 23, 25, 28–31)	8 (16, 22, 23, 25, 28–31)	8	9	8	9
	14 (2, 3, 4, 16, 22–31)	13 (2, 3, 16, 22–31)	15	13	21	27
5	12 (2, 3, 4, 5, 16, 21, 22, 23, 28–31)	11 (2, 3, 14, 15, 22, 23, 25, 28–31)	13	11	18	25
	12 (2, 3, 4, 16, 21, 22, 23, 25, 28–31)	10 (2, 3, 16, 22, 23, 25, 28–31)	13	11	18	24
	14 (2, 3, 4, 16, 22–31)	14 (2, 3, 4, 16, 22–31)	18	18	35	50
10	12 (2, 3, 4, 5, 16, 21–23, 28–31)	12 (2, 3, 4, 16, 22, 23, 25, 27–31)	16	16	34	48
	13 (2, 3, 4, 5, 14, 16, 21–23, 28–31)	13 (2, 3, 4, 14, 16, 21–23, 25, 28–31)	16	15	34	47

Bold values in columns 2 and 3 represent the number of opened stations, respectively, by P1 and P2 solutions

Table 4 Response time analysis

RK	20–15 (%)	25–20 (%)	30–25 (%)
P1_ < 10	69	62	62
P2_ < 10	68	59	58
P1_ < 15	98	90	87
P2_ < 15	98	91	85

20–15, 13 QRV and 13 BLS ambulances are required. This number of vehicles is very close to the maximal number of vehicles that can be offered by the Ministry of Health (which is 14 for both vehicles). Recall that the ambulance busy fraction is calculated based on the number of ambulances in the EMS system. Therefore, the ambulance busy fraction in the proposed configuration is also very close to the one initially used in P1 (30%). Subsequently, as expected, the EMS system configuration proposed after running P1 with this new value of busy fraction was similar to the one initially obtained. Nevertheless, if the number of ambulances deployed in the proposed EMS system configuration is quite different of the one initially used, an iterative solution approach scheme should be adopted. In iteration n ($n > 1$), P1 is solved using busy fractions $\sigma_k^{(n)}$. The number of vehicles of type k proposed by the solution at iteration n ($\sum_j Z_{jk}^{(n)}$) is determined and compared to the one obtained in the solution of iteration $n-1$. If the two solutions propose the same number of ambulances, then stop; otherwise estimate the busy fraction based on iteration n solution and run iteration $n+1$.

The comparison of the two configurations presented in Table 2, did not take into account vehicle dimensioning. For scenario 5, a comparison of (P1) and (P2) configurations while considering both, the opened vehicle stations and vehicle dimensioning, has highlighted an uphill in cost cutting of 5.64%. Obviously, one of the advantageous of considering final patient's destination in the design of EMS system remains in the additional accuracy that could be gained in term of vehicle dimensioning.

For the different scenarios, we tested problem instances while using three different standards of response time (RK). For all obtained configurations, by (P1) and (P2) models, we estimated the percentage of requests with a response time less than 10 and 15 min. Table 4 reports the average percentage of requests with a response time: (1) less than 10 min (referred to as P1_ < 10 and P2_ < 10) and; (2) less than 15 min (referred to as P1_ < 15 and P2_ < 15).

We can notice that there is a difference between the (P1) and (P2) configurations in term of service level. Nevertheless, the latter is impacted by the response time standard introduced as a parameter in each of the considered models. Obviously, when the response time parameter (RK) is higher, the service level becomes lower. On the other hand, when the response time parameter (RK) is lower, the EMS configuration cost becomes higher. When we introduce response time parameters (25–20), instead of 20–15, the EMS configuration cost is reduced, on average, for P1 and P2, respectively, by 8 and 9%. This permits to highlight the trade-off between

Table 5 Simulation results

Error level	P1_configuration		P2_configuration		AvGAP (%)	MinGAP (%)	MaxGAP (%)
	AC_P1	StdC_P1	AC_P2	StdC_P2			
10%	6,291,374	46,719	6,493,490	51,700	3.31	2.89	3.81
20%	6,402,564	76,187	6,662,675	84,519	3.56	2.86	4.92
30%	6,342,385	97,681	6,561,712	100,423	3.46	2.19	4.82

the cost incurred by the EMS system and the service level. Improving the service level leads to increase the investment in the EMS system.

In this section, we intend to compare the two approaches while taking into account the quality of demand forecasts. Indeed, in the previous comparison of the two models, we have not considered the error in demand forecasts and how it can impact the performance of the EMS system. Herein, we consider that no error is made on the set of hospitals serving the requests of each demand zone; however, an error could be made on the number of requests taken over by each of these hospitals. Three error levels have been considered: 10%, 20% and 30%. For each error level, the demand is generated while using a discrete uniform distribution. The impact of the demand forecast error is assessed while considering scenario 5 with a response time standard parameter (20–15). For each error level, twenty independent runs are conducted on P1 while considering the generated demand, and, the opened sites, as per (P1) and (P2) solutions. In Table 5, the following information is provided: average and standard deviation of the EMS system cost obtained by (P1) and (P2) configurations (AC_P1, StdC_P1, AC_P2, StdC_P2) and, the average, the minimum and the maximum gap (AvGAP, MinGAP, MaxGAP), between the two configurations as stated above.

A cost cutting is achieved for all forecast error levels. With a higher error level, the GAP range becomes wider. Nevertheless, even with an error level of 30%, the proposed approach remains advantageous, as the minimum cost cutting obtained through the use of the proposed approach is 2.19%. In case we know with certainty the hospitals serving each demand zone, it is beneficial to consider the proposed approach as it permits a cost cutting besides a more accurate vehicle dimensioning, avoiding overestimation of capacity requirements of this resource.

5 Conclusions

In this paper, we addressed the design of the EMS system. A novel approach based on the consideration of patient's final destination, is proposed. This approach is compared to its counterpart approach that does not consider this kind of information. Two mixed integer linear programs are developed. Experimentation of the two models was conducted on 10 different demand scenarios based on a real life case and pointed out the usefulness of the proposed approach. On average, a cost saving of 3% is achieved. This cost saving is doubled for scenarios with a demand coefficient

of 5, 6 and 7. Moreover, cost saving is accompanied by a reduction in vehicle round-trip time that implies an increase in BLS vehicle availability. The proposed approach presents the advantage of providing a more accurate BLS vehicle dimensioning as it avoids overestimation of capacity requirements of this resource. In case we know with certainty the hospitals serving each demand zone, this approach remains interesting even with an error level of 30% on demand forecasts.

The focus of this paper was rather placed on assessing the interest of considering vehicle round-trip information in the design of the EMS system. In practice, it is common that a BLS ambulance can be considered available just after transferring patients to hospital. Indeed, when there is no risk of contamination or if ambulance cleaning and disinfection can take place in the hospital just after patient drop-off, the ambulance can serve an emergency request without necessarily returning to its base station. In this case, the same service level can be ensured while deploying a lower number of ambulances.

References

- Beraldi P, Bruni ME (2009) A probabilistic model applied to emergency service vehicle location. *Eur J Oper Res* 196:323–331
- Beraldi P, Bruni ME, Conforti D (2004) Designing robust emergency medical service via stochastic programming. *Eur J Oper Res* 158:183–193
- Basar A, Catay B, Ünlüyurt T (2011) A multi-period double coverage approach for locating the emergency medical service stations in Istanbul. *J Oper Res Soc* 62(4):627–637
- Boujemaa R, Hammami S, Jebali A (2013) A stochastic programming model for ambulance location allocation problem in the Tunisian context. In: International conference on industrial engineering and systems management, IESM'2013, October 28–30, RABAT—MOROCCO
- Boujemaa R, Jebali A, Hammami S, Ruiz A, Bouchriha H (2018) A stochastic approach for designing two-tiered emergency medical service systems. *Flex Serv Manuf J* 30(1–2):123–152
- Church RL, ReVelle CS (1974) The maximal covering location problem. *Pap Reg Sci Assoc* 32:101–118
- Correia I, Captivo ME (2003) A Lagrangean heuristic for a modular capacitated location problem. *Ann Oper Res* 122(1):141–161
- Daskin MS (1982) Application of an expected covering model to emergency medical service system design. *Decis Sci* 13:416–439
- Daskin MS (1983) A maximum expected location problem: formulation, properties and heuristic solution. *Transp Sci* 17:416–439
- Daskin MS, Stern EH (1981) A hierarchical objective set covering model for emergency medical service vehicle deployment. *Transp Sci* 15:137–152
- Doerner KF, Gutjahr WJ, Hartl RF, Karall M, Reimann M (2005) Heuristic solution of an extended double-coverage ambulance location problem for Austria. *Central Eur J Oper Res* 13(4):325–340
- Gendreau M, Laporte G, Semet F (1997) Solving an ambulance location model by tabu search. *Locat Sci* 5:75–88
- Hogan K, ReVelle CS (1986) Concepts and application of backup coverage. *Manag Sci* 34:1434–1444
- Ingolfsson A, Erkut E, Buge S (2003) Simulating a single start station for edmonton EMS. *J Oper Res Soc* 54:736–746
- Ingolfsson A, Budge S, Erkut E (2008) Optimal ambulance location with random delays and travel times. *Health Care Manag Sci* 11:262–274
- Jebali A, Hammami S, Boujemaa R (2012) A mathematical model for ambulance location-allocation problem in the Tunisian context. In: International conference on computer related knowledge, ICCRK 2012, 5–7 July 2012, Sousse, TUNISIE
- Larson RC (1974) A hypercube queueing model for facility location and redistricting in urban emergency service. *Comput Oper Res* 1:67–95

- López B, Innocenti B, Busquets D (2008) A multiagent system for coordinating ambulances for emergency medical services. *IEEE Intell Syst* 23(5):50–57
- Maleki M, Majlesinasab N, Sepehri MM (2014) Two new models for redeployment of ambulances. *Comput Ind Eng* 78:271–284
- Mandell MB (1998) Covering models for two-tiered emergency medical services system. *Locat Sci* 6:355–368
- Marianov V, ReVelle CS (1992) A probabilistic fire-protection siting model with joint vehicle reliability requirements. *Pap Reg Sci* 71:217–241
- McLay LA (2009) A maximum expected covering location model with two types of servers. *IIE Trans* 41(8):730–741
- Noyan N (2010) Alternate risk measures for emergency medical service system design. *Ann Oper Res* 181:559–589
- Revelle C, Hogan K (1988) A reliability-constrained siting model with local estimates of busy fractions. *Environ Plann B: Plann Des* 15:143–152
- Patel AB, Waters NM, Blanchard IE, Doig CJ, Ghali WA (2012) A validation of ground ambulance pre-hospital times modeled using geographic information systems. *Int J Health Geographics* 11(1):42
- Schilling DA, Elzinga DJ, Cohon J, Church RL, ReVelle CS (1979) The TEAM/FLEET models for simultaneous facility and equipment setting. *Transp Sci* 13:163–175
- Silva PMS, Pinto LR (2010) Emergency medical systems analysis by simulation and optimization. In: Simulation conference (wsc), proceedings of the 2010 Winter, pp 2422–2432. <https://doi.org/10.1109/wsc.2010.5678938>
- Silva F, Serra D (2008) Locating emergency services with different priorities: the priority queuing covering location problem. *J Oper Res Soc* 59:1229–1238
- Sonoda T, Ishibai K (2015) Project on Information-support solution in emergency medical system. *Fujitsu Sci Technol J* 51(3):39–49
- Su S, Shih CL (2003) Modeling an emergency medical services system using computer simulation. *Int J Med Inform* 72:57–72
- Sudtachat (2014) Strategies to improve the efficiency of emergency medical service (EMS) systems under more realistic conditions. Doctorat Thesis, Clemson University
- Schmid V (2012) Solving the dynamic ambulance relocation and dispatching problem using approximate dynamic programming. *Eur J Oper Res* 16(3):611–621
- Sudlow A, McConnell N, Egan G, Jansen JO (2011) Destination healthcare facility of patients with suspected traumatic brain injury in Scotland: Analysis of pre-hospital data. *Injury* 44(9):1237–1240
- Takeda RA, Widmer JA, Morabito R (2007) Analysis of ambulance decentralization in urban emergency medical service using the hypercube queueing model. *Comput Oper Res* 34(3):727–741
- Tien JM, EL-Tell K, Simons GR (1983) Improved formulations to the hierarchical health facility location-allocation problem. *IEEE Trans Syst Man Cybern* 13(6):1128–1132
- Toregas C, Swain R, ReVelle CS, Bergman L (1971) The location of emergency service facilities. *Oper Res* 19:1363–1373
- Van Essen JT, Hurink JL, Nickel S, Reuter M (2014) Models for ambulance planning on the strategic and the tactical level. Beta Publishing
- Yin P, Mu L (2012) Modular capacitated maximal covering location problem for the optimal siting of emergency vehicles. *Appl Geogr* 34:247–254
- Zhang Z, Jiang H (2014) A robust counterpart approach to the bi-objective medical service design problem. *Appl Math Model* 38:1033–1040
- Zuidhof GM (2010) Capacity planning of ambulance services: statistical analysis, forecasting and staffing. Dissertation, Amsterdam University

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